

Proposal 1: GitHub Repository Explorer – Intelligent Repository Analysis Platform

1. Introduction

The rapid growth of open-source development has resulted in millions of repositories on GitHub, making it difficult to identify reliable and well-maintained projects. Popularity indicators such as stars and forks often fail to reflect true project quality and sustainability.

The GitHub Repository Explorer is a web-based intelligent platform designed to discover, analyze, and compare repositories using real-time GitHub API data. Instead of relying on superficial metrics, the system evaluates contributor activity, commit frequency, issue resolution rates, pull request trends, and overall repository health. It may also include AI-driven recommendations to provide personalized insights.

2. Problem Statement

Despite available statistics on GitHub, evaluating repository quality remains inefficient.

- Developers struggle to find active and trustworthy projects.
- Students have difficulty identifying suitable repositories for learning.
- Recruiters spend excessive time reviewing GitHub activity manually.
- Inactive repositories may still appear popular.
- No unified system converts repository data into clear health indicators.

This leads to wasted time, poor decision-making, and reduced productivity.

3. Objectives

General Objective:

To develop a scalable platform that evaluates and compares GitHub repositories using data-driven metrics and intelligent analysis.

Specific Objectives:

- Integrate GitHub REST and GraphQL APIs.
- Implement repository health metrics (commit consistency, issue closure rate, pull request merge ratio, contributor growth).
- Develop a comparison engine and interactive analytics dashboard.
- Implement an AI-based recommendation system.
- Create a ranking mechanism using weighted performance indicators.

4. Key Features

- Repository discovery with filtering options.
- Dynamic repository health scoring.
- Contributor analytics and trend visualization.
- Side-by-side repository comparison.
- Optional AI-powered recommendations.

5. Significance and Impact

Academically, the project demonstrates API integration, backend architecture, data analytics, visualization, scalability, and AI integration.

Practically, it enables developers, students, recruiters, and organizations to make faster and more informed decisions.

By transforming raw GitHub data into structured insights, the GitHub Repository Explorer improves productivity and introduces a data-driven approach to open-source evaluation.

Proposal 2: Telemedicine Platform – A Doctor–Patient Telehealth System

1. Introduction

The rapid growth of digital technology offers a strong opportunity to improve healthcare delivery in Ethiopia, especially for rural and underserved communities. This project proposes a **Telemedicine Platform**, a web-based system designed to directly connect patients with **licensed doctors** through secure video consultations, appointment scheduling, and electronic prescription services. By focusing on direct doctor–patient interaction, the system maintains a manageable scope while still addressing major healthcare accessibility challenges. The platform will also integrate local Ethiopian payment methods, making it practical and usable in real-world settings. From a software engineering perspective, the project emphasizes real-time communication, secure system design, and responsive user interfaces, resulting in a realistic and implementable final-year project.

2. Statement of the Problem

Ethiopia's healthcare system faces significant challenges related to access, efficiency, and availability of medical professionals. Doctors are primarily concentrated in urban areas, forcing many patients—especially those in rural regions—to travel long distances and incur high costs for basic medical consultations. Additionally, traditional appointment systems rely on physical queues or phone calls, leading to long waiting times, overcrowded facilities, and inefficient use of doctors' time. Patients also have limited visibility into doctor availability and specialties, making it difficult to find appropriate medical care in a timely manner.

3. Objectives

General Objective:

To design and develop a secure, scalable telemedicine platform that enables direct digital consultation between patients and doctors, improving healthcare accessibility and efficiency across Ethiopia.

Specific Objectives:

- To implement secure real-time video consultations using WebRTC technology with encryption to protect patient privacy.
- To develop an appointment scheduling system that allows doctors to set availability and patients to book consultations based on specialty and time slots.
- To design an electronic prescription module that enables doctors to generate and share digital prescriptions with patients.
- To integrate local payment methods such as chapa/Telebirr to support consultation fee payments and receipt generation.
- To implement basic emergency consultation features that allow urgent cases to receive priority access to available doctors.

4. Significance of the Project

This project provides strong academic and practical value. Academically, it allows the development team to apply core software engineering concepts such as real-time communication, secure authentication, database design, and payment gateway integration within a realistic scope. Practically, the platform improves access to qualified doctors for patients in remote areas, reduces unnecessary travel and overcrowding in hospitals, and enables faster medical guidance.

Proposal 3: Inventory Management System for Small Shops

1. Introduction

Small retail shops in Ethiopia often rely on manual record-keeping methods such as notebooks and spreadsheets to manage inventory. These approaches frequently result in stock shortages, overstocking, and inaccurate records. This project proposes an **Inventory Management System** that automates product tracking, stock control, sales recording, and report generation. The system aims to provide shop owners with real-time visibility into their inventory and business performance. From a software engineering perspective, the project involves designing efficient database architectures, implementing real-time stock monitoring, enforcing role-based access control, and creating intuitive user interfaces, resulting in a practical and production-ready business management solution.

2. Statement of the Problem

Small retail shops in Ethiopia face significant operational challenges due to ineffective inventory management practices. Manual record-keeping using handwritten notebooks and spreadsheets often leads to frequent errors in tracking stock movements and sales, producing unreliable inventory data. Shop owners lack real-time visibility into current stock levels, which results in unexpected stock-outs and lost sales opportunities. In addition, the absence of accurate reports limits informed financial decision-making, making it difficult to identify best-selling products, slow-moving items, or profitable pricing strategies. Manual systems also fail to properly track losses due to theft, damage, or expiration, leading to unexplained inventory discrepancies that reduce profit margins.

3. Objectives

General Objective:

To design, develop, and deploy an inventory management system that enables small retail shops to efficiently manage products, monitor stock levels in real time, and make data-driven business decisions.

Specific Objectives:

- To develop comprehensive product management features that allow shop owners to register and manage products with details such as name, category, price, and reorder levels, including search and filtering functionality.
- To implement automated stock tracking through transaction-based updates for sales and restocking activities, ensuring accurate inventory levels and complete audit trails.
- To enforce data integrity and business rules by preventing negative stock values, duplicate entries, and inconsistent pricing.
- To create reporting and analytics modules that generate sales summaries, stock valuation, profit margins, and product performance reports with customizable date ranges.
- To implement proactive low-stock alert mechanisms that notify shop owners when inventory falls below defined reorder thresholds.
- To establish role-based access control with defined user roles such as Owner, Manager, and Staff, each with appropriate system permissions.

4. Significance of the Project

This project offers value both academically and practically. From a software engineering perspective, it strengthens skills in database design, transaction management, authentication, authorization, and user-centered interface design. From a practical standpoint, the system reduces manual errors, provides real-time visibility into business operations, and supports data-driven decision-making. It minimizes financial losses caused by stock-outs and overstocking, improves customer satisfaction through better product availability.